Quieting Food Noise After GLP-1 Discontinuation: A Feasibility Study of a Novel Mindful Eating Program

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PURPOSE

Stopping GLP-1s Creates Behavioral Chaos — and No One Has a Plan for It

- GLP-1 receptor agonists (semaglutide and tirzepatide) are blockbuster drugs that reduce appetite, cravings, "food noise," and body weight.
- One in eight adults (12%) have taken one [1].
- 54% of users discontinue by 1 year; 72% within 2 years AND 46-57% restart within 2 years [2].
- One year after stopping semaglutide, participants regained 2/3 of lost weight and had return of cardiometabolic risk factors [3].

By targeting food noise as an upstream cognitive and behavioral mechanism implicated in problematic eating, this study focuses on a modifiable driver of eating behavior.

The findings are intended to inform future interventions aimed at reducing the risk of returning to pre-medication behaviors following GLP-1 discontinuation.

BACKGROUND & RATIONALE

- GLP-1 receptor agonists (semaglutide and tirzepatide) are widely prescribed medications that are increasingly used for obesity treatment and are effective in reducing appetite and food-related intrusive thoughts, commonly referred to as "food noise." [4]
- However, many individuals discontinue GLP-1 therapy due to cost, side effects, access limitations, or personal preference.
- After cessation of weight management medications (WMMs), body weight was regained at an average rate of 0.4 kg per month, with a projected return to baseline weight by 1.7 years [5]

Following discontinuation, patients frequently experience the return of food noise, dysregulated eating patterns, and weight regain, highlighting the need for clinically viable strategies that clinicians can offer to support patients during planned or necessary transitions off pharmacologic therapy.

- There are currently no evidence-based behavioral interventions designed to function as a clinically meaningful off-ramp following GLP-1 therapy, limiting clinicians' ability to fully support patients' eating behavior and self-regulation needs following medication discontinuation.
- Mindfulness-based eating programs have demonstrated promise in improving interoceptive awareness, attentional control, and reducing rumination—mechanisms implicated in food noise. [6]
- Targeting food noise as an upstream cognitive and attentional mechanism may help prevent re-emergence of dysregulated eating behaviors after medication cessation.

This scalable, skills-based approach provides structured support for individuals transitioning off GLP-1 therapy and for those who cannot remain on pharmacologic treatment.

AIMS

This study evaluates whether a structured mindful eating intervention is feasible and effective for individuals discontinuing GLP-1 therapy.

Aim 1 – Feasibility

Evaluate recruitment success, retention, adherence, and participant satisfaction of a 12-week online mindful eating program delivered to adults who have discontinued GLP-1 therapy.

Aim 2 – Preliminary Efficacy

Examine changes in:

- Food noise (RAID-FN Food Noise Inventory) [8]
- Eating-related distress
- Anxiety and depression symptoms
- Quality of life

Aim 3 – Durability & Behavioral Adaptation

Assess sustainability of effects at 3- and 6-month follow-up and explore:

- Behavioral adaptation following medication cessation
- Rates of GLP-1 re-initiation

CONCEPTURAL FRAMEWORK

- GLP-1 receptor agonists primarily quiet physiology by reducing appetite signaling, dampening reward salience, and decreasing intrusive food-related thoughts ("food noise"). However, they do not build durable self-regulation skills.
- When medication is discontinued, individuals may enter a neurobehavioral vulnerability window in which food-related rumination, cue reactivity, and emotional eating re-emerge. Without regulatory skills in place, many patients re-enter a chaotic eating cycle characterized by overeating, subsequent restriction, renewed food preoccupation, and repeated loss of perceived control.
- We conceptualize this pattern as behavioral chaos (**Figure 1**). Pharmacologic appetite suppression is removed, but the cognitive and attentional systems required for adaptive regulation remain underdeveloped. As shown in **Table 1**, GLP-1 medications and mindful eating operate on complementary regulatory systems. GLP-1s reduce bottom-up biological drivers of appetite and reward during treatment. In contrast, mindful eating strengthens top-down processes — attentional control, interoceptive awareness, and flexible decision-making — that are essential once pharmacologic support is withdrawn.
- The proposed intervention applies the Am I Hungry?® Mindful Eating Cycle (**Figure 2**) [7] as a structured exit strategy. Rather than reacting automatically to intrusive thoughts or external cues, participants learn to pause and move intentionally through the cycle: *Why am I eating? When do I want to eat? What do I eat? How do I eat? How much do I eat? Where do I invest my energy?* By repeatedly engaging this reflective process, individuals interrupt the **overeat → restrict → repeat** loop and replace chaotic reactivity with internally guided regulation.
- Thus, GLP-1 therapy and mindful eating are complementary across time. Medications reduce symptom burden during active treatment; mindful eating builds the durable cognitive and behavioral capacity needed to sustain stability after discontinuation.

METHODS

Study Design

Randomized controlled feasibility trial with:

- Intervention group (immediate mindful eating program)
- Wait-list control group (usual care during initial period, then crossover)

Assessments conducted at:

- Baseline
- Post-intervention (12 weeks)
- 3-month follow-up
- 6-month follow-up

Participants Inclusion Criteria:

- Adults who have discontinued GLP-1 receptor agonist therapy within 2-3 weeks
- No diagnosis of diabetes
- No active eating disorder
- Interest in improving relationship with food

Recruitment:

- National recruitment via social media and online platforms

Intervention

Delivered in a format similar to the Am I Hungry?® Mindful Eating Program:

- 12 weekly sessions
- 90-minute online group workshops
- Led by a trained facilitator

Outcome Measures

Primary Feasibility Outcomes

- Recruitment rate
- Retention rate
- Session attendance/adherence
- Participant satisfaction

Preliminary Efficacy Outcomes

- Food noise (RAID-FN Food Noise Inventory) [8]
- Eating-related distress
- Anxiety (validated scale)
- Depression (validated scale)
- Quality of life

Exploratory Outcomes

- Behavioral adaptation following GLP-1 cessation
- GLP-1 re-initiation

Figure 1. Chaos → Calm Adaptive Pathway Model

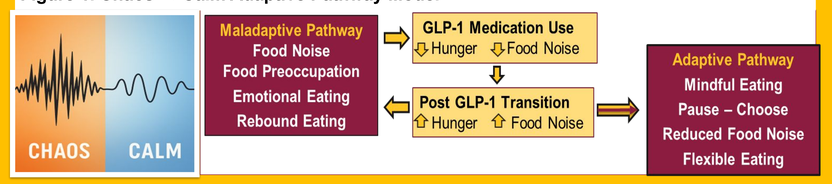
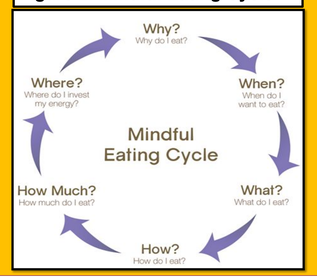


Table 1.
Complementarity: Pharmacologic vs. Behavioral Regulation

GLP-1 Medications	Mindful Eating Intervention
Quiet bottom-up appetite signaling	Strengthen top-down self-regulation
Reduce reward salience	Improve attentional control
Suppress physiological hunger	Restore interoceptive awareness
Reduce food noise pharmacologically	Reduce food noise cognitively
Provide symptom relief	Build durable regulatory capacity

Figure 2. Mindful Eating Cycle



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